Institute of Mathematical Sciences
University of Malaya

SIM1001 Basic Mathematics
SIX1013 Fundamental of Advanced Mathematics

TUTORIAL 3 (Part 1)

1. Given the universal set $U = \{x \in \mathbb{Z} \mid 0 \le x \le 20\}$ and
 $A = \{ x \mid x$ is a multiple of 3$\}$, $B = \{ x \mid x$ is prime$\}$
 $C = \{ x \mid x$ has exactly two positive factors$\}$, $D = \{x \mid x(x^2-49)(x^2-9)(x^2-1)=0\}$
 $E = \{ x \in A \mid x$ is even$\}$.
 List the elements in each following sets.

(a)	$(A \cup B)'$	(b)	$(A \cup B)' \cap D$
(c)	$(B \cap C)'$	(d)	$C' \cap (D \cup E)$
(e)	$(C - D) \cup E$	(f)	$(A \cup D) - (B \cup E)$

2. Are the sets A and B equal?
 (a) $A = \{ x \mid x = 4n,\ n \in \mathbb{Z} \}$, $B = \{ x \mid x = 2n$ where n is even$\}$

 (b) $A = \{ x \mid x = 2n + 1,\ n \in \mathbb{Z} \}$, $B = \{ x \mid x = 2n - 9,\ n \in \mathbb{Z} \}$,

 (c) $A = \{ x \mid x = 4n + 1,\ n \in \mathbb{Z} \}$, $B = \{ x \mid x = 2n + 1$ where n is odd$\}$

 (d) $A = \{ x \in \mathbb{R} \mid x^2 + 2 = 0\}$, $B = \{ x \in \mathbb{Q} \mid x^2 - 2 = 0\}$

3. Let $A = \{1, 2, 3, 6\}$, $B = \{1, 4, 5\}$.
 Compare $(A \times B) \cap (B \times A)$ with $(A \cap B)^2$.

4. Let R be the relation on $A = \{x \in \mathbb{N} \mid x$ is a factor of 18$\}$ and $x\,R\,y \Leftrightarrow x + 3y = 12$.

 (a) Write R as a set of ordered pairs.
 (b) Find the matrix of R.
 (c) Find Domain of R and Range of R.
 (d) Find R^{-1}

5. Let R_i, $i = 1, 2, 3, 4$ be relation on the set $A = \{a, b, c, d\}$.
 Given that $R_1 = \{(a, b), (d, c), (b, b), (b, a), (c, a)\}$,
 $R_2 = \{(b, c)\}$,
 $R_3 = \{(b, b), (b, c), (c, b), (c, c)\}$ and
 $R_4 = \{(a, a), (b, b), (c, c), (d, d)\}$.
 (a) For each R_i, $i = 1, 2, 3, 4$, determine whether or not the relation is
 (i) reflexive (ii) symmetric (iii) transitive.

 (b) Is R_i, $i = 1, 2, 3, 4$ a function from A into A? If so, determine its image.

6. Determine whether or not, the relation R on the set A is an equivalence relation.

(a) $A = \mathbb{R}$, $x\,R\,y \Leftrightarrow x \le y$.

(b) $A = \{$straight lines in the $xy-$plane$\}$, $x\,R\,y \Leftrightarrow x$ is parallel to y.

(c) $A = \mathbb{N}$, $x\,R\,y \Leftrightarrow y$ is a multiple of x.

7. Let $A = \{-1, 0, 1\}$, $B = \left\{0, 1, \frac{3}{5}\right\}$, $C = \{-2, 0, 3, 6\}$ and $D = \left\{\frac{1}{2}, 1, 2\right\}$.

Let f, g, h, k be functions defined as follows :

$f: A \to B,\ f(x) = x^2$, $\qquad$ $g : B \to C,\ g(x) = 3 - 5x$,

$h : C \to D,\ h(x) = \dfrac{|x-2|}{2}$ and $k : D \to A,\ k(x) = \cos \pi x$.

(a) Which of the above functions are one-to-one?

(b) Which of the above functions are onto?

(c) Find $f \circ k$, $g \circ f$, $h \circ g$, $h \circ g \circ f$ and $k \circ h \circ g \circ f$

8. Identify those functions in (a) – (f) which are (i) one-to-one (ii) onto (iii) one-to-one correspondence. In each case, identify the image set (range) of the function.

(a) $f_1 : \{1, 2, 3, 4, 5\} \to \{2, 3, 4, 5, 6\}$ be the function defined by $f_1(x) = x + 1$.

(b) $f_2 : \mathbb{R} \to \mathbb{R}$ be the function defined by $f_2(x) = x^2 - 3$.

(c) $f_3 : \mathbb{Z} \to \mathbb{Z}$ be the function defined by $f_3(x) = x^2 + 1$.

(d) $f_4 : \mathbb{R} \to \mathbb{R}$ be the function defined by $f_4(x) = x^3 - 5$.

(e) $f_5 : \mathbb{N} \to \mathbb{N}$ be the function defined by $f_5(x) = \begin{cases} x+1, & x \text{ is odd} \\ x-1, & x \text{ is even} \end{cases}$.

(f) $f_6 : \mathbb{R} \to \mathbb{R}$ be the function defined by $f_6(x) = \begin{cases} x^2+1, & x \ge 0 \\ 5x-3, & x < 0 \end{cases}$.

Which of the functions in (a)-(f) are invertible? For each invertible function in (a)-(f), find its inverse.

9. Let $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ be defined as follows :

$f(x) = x^2 + 4x - 1$ and $g(x) = 2x + 5$.

Find (a) $(f \circ g)\,(x+1)$ $\qquad$ (b) $(g \circ f)\,(3x)$

(c) $(f \circ f)\,(-x)$ $\qquad$ (d) $(g \circ g)\,(x^2 - 1)$

****